Use Case: Validate Move

Iteration: 1

Primary Actor: System

Goal in Context: Ensure that a player's move is valid according to the rules of the game.

Preconditions:

1. The player is logged into their account.
2. The player is connected to an active game session.
3. It is a player's turn to make a move.
4. The player has selected a specific grid location on the game board.
5. The selected move complies with the basic rules of the game (e.g., selecting an empty cell in Tic-Tac-Toe or a valid jump in Checkers).

Trigger: A player attempts to place a piece or make a move on the board.

Scenario:

1. A player selects a location on the board to place a piece or move a piece.
2. The system verifies that it is the player's turn.
3. The system checks that the selected move follows the specific game rules:
    a. Tic-Tac-Toe:
        i. The system ensures the selected cell is unoccupied.
    b. Checkers:
        i. The system confirms that the move is within the piece's movement range (diagonal move for regular pieces, any direction for kings).
        ii. The system checks if the move is a valid capture, if applicable.
    c. Connect 4:
        i. The system verifies that the selected column has an open spot in the lowest available row.
4. If the move is valid, the system allows the move to proceed.
5. If the move is invalid, the system displays an error message and prompts the player to select a valid move.

Postconditions:

1. The move is validated and, if valid, will be executed on the board.
2. If the move is invalid, the system prevents the action and requests the player to make another selection.

Exceptions:

- A player attempts to place a piece outside the boundaries of the board.
- A player selects an occupied spot (e.g., in Tic-Tac-Toe).
- A player makes an invalid move for the selected piece (e.g., in Checkers, pieces can only move diagonally on dark squares).

Priority: High, as move validation is essential to prevent invalid actions in each game.

When Available: 1st iteration.

Frequency of Use: Multiple times per game, depending on game length.

Channel to Actor: The system automatically checks and notifies players via the game interface.

Secondary Actors: Both players in the game session.

Channels to Secondary Actors: If the move is valid, the updated board state is displayed on each player’s screen

Open Issues: None.